Kasi Yedumati · Full Stack Training

# Session Calendar — 21 Weeks × 105 Live Sessions

**Schedule source of truth.** Follow the modules listed here in order. Classroom pacing follows this calendar.

| Setting | Value |
|---|---|
| Live class | **1 hour 15 minutes** |
| Days | Mon–Fri (5 sessions / week) |
| Length | **21 weeks** (20 + bridge week 12A for HTML & CSS) |
| Total live sessions | **105** (S1–S100 + HTML & CSS sessions F1–F9, which replace the old S59–S62) |
| Live hours | 105 × 1.25h = **131.25 hours** |
| Homework | ~45–90 min after most sessions |

**Compressed option:** 16 weeks is *not* recommended for this syllabus at 75 minutes/day. Use only with Saturday labs or fewer React/JPA outcomes.

**Rule:** One live session = **one primary focus** (one concept block + one mini-lab). Do not schedule two full modules in one 75-minute class.

**Engagement rule:** Start from a lab ticket ( `15_Lab_Assets/` , or `GIT/labs/` , `FRONTEND/HTML_CSS/labs/` , `FRONTEND/labs/` , `FRONTEND/REACT/labs/` ), then open module notes for Definition/Purpose as needed.

**Biweekly Assessment:** After every even week (Weeks 2, 4, … 20), complete the matching paper from `16_Assessments/` (BW01–BW10) as homework or Friday wrap-up. Graded papers live only under `16_Assessments/` — not inside modules.

**Shared topics (repo root):** `GIT/` · `MYSQL/` · `FRONTEND/` (includes `FRONTEND/REACT/` ) — scheduled here, authored outside this folder.

**Typical class shape** (timing lives here — not repeated in every module):

| Minutes | Block |
|---:|---|
| 5 | Open lab kit |
| 25 | Concept demo / notes |
| 35 | Lab (starter/broken) |
| 10 | Exit + homework |

## How to read the table

| Column | Meaning |
|---|---|
| **S#** | Global session number (01–100) |
| **Focus** | What you practice *in class* |
| **Module file** | Notes to open |
| **Homework** | Must finish before next session |

**Module file:** click the link (or keep **same** = previous session's file).

**Authoring status:** Foundations + Core Java + OOP Parts 1–3 + Collections + Exception Handling + File Handling + Git/GitHub + MySQL bridge + JDBC CRUD + Maven/Spring Boot setup (`01`) + Spring Web REST (`02`) + Spring Data JPA (`03`) + REST DTOs / Postman (`04`) + Spring Security lite (`05`), lab kits 01–19 (kit 19: JWT register/login + protect POST; Lab 06 freeze; Lab 07 Actuator/health + env/PaaS/M0), Servlet vs Spring interview map (`08`, kit 19 point-only), and BW01–BW06 exist. Shared FRONTEND HTML/CSS/JS is already used. Protected labs `06` (CRUD + auth freeze) and `07` (deploy debug, kit 19) are authored. Later phases (CORS recap via `04`, React, testing, capstone M1–M3) are still planned names until authored.

## Phase A — Foundations & Core Java (Sessions 01–30)

| S# | Wk | Focus | Module file | Homework |
|---|---|---|---|---|
| 01 | 1 | Computer, OS, files, terminal basics | 01_Foundations/01_Computer_and_Programming_Foundations.md | Workspace folders |
| 02 | 1 | Client–server + roles; VS Code / IntelliJ tour | same | Roadmap note |
| 03 | 1 | Install JDK 17; verify PATH, `javac` / `java` | 02_Core_Java/01_Java_Setup_and_First_Programs.md | Setup proof screenshot |
| 04 | 1 | `public class` + `main`; first programs | same | `HelloJava` on GitHub account created |
| 05 | 1 | Keywords, identifiers; lab catch-up | same | Flash keywords |
| 06 | 2 | Variables & primitive types | 02_Core_Java/02_Variables_Datatypes_Operators_IO.md | Type drills |
| 07 | 2 | Operators | same | Calculator v1 |
| 08 | 2 | `Scanner` I/O & printing | same | Canteen calculator + receipt print |
| 09 | 2 | Casting & wrappers lite | same | Fix 5 casting mistakes |
| 10 | 2 | Mini: CLI calculator polish | same | Push repo |
| 11 | 3 | `if` / `else` | 02_Core_Java/03_Flow_Control.md | Grade calculator |

| S# | Wk | Focus | Module file | Homework |
|---|---|---|---|---|
| 12 | 3 | `switch` | same | Menu with switch |
| 13 | 3 | `for` / `while`, break/continue | same | Tables & number guessing |
| 14 | 3 | Nested menus | same | Start ATM menu |
| 15 | 3 | ATM menu lab day | same | Finish ATM + push |
| 16 | 4 | Strings: immutability, index, `substring` | 02_Core_Java/04_Strings.md | Slice drills |
| 17 | 4 | String methods; `equals` vs `==` | same | Email/phone parse |
| 18 | 4 | Arrays 1D + enhanced `for` | 02_Core_Java/05_Arrays.md | Marks array |
| 19 | 4 | 2D arrays + simple algos | same | Max / search lab |
| 20 | 4 | Marks Manager mini-project | same | Push CLI app |
| 21 | 5 | Static methods, return, params | 02_Core_Java/06_Methods.md | Extract calculator methods |
| 22 | 5 | Method overloading | same | Overload drills |
| 23 | 5 | Refactor ATM into methods (one class) | same | Methods-only ATM |
| 24 | 5 | Class / object / `new` | 03_OOP_Java/03_OOP_Part1_Classes_Encapsulation.md | Student class |
| 25 | 5 | Constructors + `this` | same | Constructor drills |
| 26 | 6 | Instance methods vs static | same | Move logic onto objects |
| 27 | 6 | Encapsulation (`private`, getters) | same | Hide balance / marks |
| 28 | 6 | Instance vs static fields | same | Counter vs per-object data |
| 29 | 6 | BankAccount lab | same | Deposit / withdraw with `print` + `return` |
| 30 | 6 | Part 1 catch-up / push | same | Push OOP Part 1 |

## Phase B — OOP, Collections, IO & Git (Sessions 31–45)

| S# | Wk | Focus | Module file | Homework |
|---|---|---|---|---|
| 31 | 7 | Inheritance + `super()` | 03_OOP_Java/03_OOP_Part2_Inheritance.md | Employee hierarchy start |
| 32 | 7 | Method overriding | same | Override `display` / `toString` lite |

| S# | Wk | Focus | Module file | Homework |
|---|---|---|---|---|
| 33 | 7 | IS-A; single class inheritance | same | Sketch campus roles |
| 34 | 7 | Abstract classes | 03_OOP_Java/03_OOP_Part3_Polymorphism_Interfaces.md | Abstract `Account` |
| 35 | 7 | Interfaces (`implements`) | same | `Payable` / `Printable` |
| 36 | 8 | Polymorphism + packages; OOP mini-project push | same | Push Campus Wallet |
| 37 | 8 | `ArrayList` / `List` | 04_Collections_Exceptions_IO/01_Collections.md | List CRUD |
| 38 | 8 | `HashSet` / uniqueness | same | Dedup emails |
| 39 | 8 | `HashMap` / lookups | same | Contact map |
| 40 | 8 | Collections mini-project | same | Push inventory / contacts |
| 41 | 9 | `try` / `catch` / `finally`; try-with-resources | 04_Collections_Exceptions_IO/02_Exception_Handling.md | Safe `Scanner` lab |
| 42 | 9 | `throw` + custom exceptions | same | Domain errors |
| 43 | 9 | Files (`java.nio` / streams) + CSV/text | 04_Collections_Exceptions_IO/03_File_Handling.md | CSV append |
| 44 | 9 | Git Session 1: first GitHub push | ../GIT/01_Git_Session1.md | KSR_MART on GitHub |
| 45 | 9 | Git Session 2: daily loop + clone | ../GIT/02_Git_Session2.md | Hygiene lab; homework Session 3 · 4 · 5 |

## Phase C — MySQL & JDBC (Sessions 46–58)

*External dependency: repo `MYSQL/` (and `FRONTEND/` / `FRONTEND/REACT/` on later weeks).*

| S# | Wk | Focus | Module / lab | Homework |
|---|---|---|---|---|
| 46 | 10 | Why DB; install/connect | MYSQL README + 00_MySQL_Setup + FS bridge | `training_db` created |
| 47 | 10 | Tables, PK; DDL CREATE | MYSQL/01–02 | 3 tables |
| 48 | 10 | DML INSERT/UPDATE/DELETE | MYSQL/02 | Seed data |
| 49 | 10 | DQL WHERE/ORDER/LIMIT | MYSQL/03 + labs seed | Report queries |

| S# | Wk | Focus | Module / lab | Homework |
|---|---|---|---|---|
| 50 | 10 | FS bridge concepts + ER sketch | 06_Database_MySQL/01_MySQL_for_Full_Stack_Developers.md | Library ER |
| 51 | 11 | Modeling PK/FK | MYSQL/06 | Related tables |
| 52 | 11 | INNER JOIN | MYSQL/07 | Join drills |
| 53 | 11 | LEFT JOIN + multi-table | MYSQL/07 | Customers w/o orders |
| 54 | 11 | TCL + indexes lite | MYSQL/09 | Rollback demo |
| 55 | 11 | JDBC: `Connection`, try-with-resources | 07_JDBC/01_JDBC_CRUD.md | Connect proof (no Spring) |
| 56 | 12 | `PreparedStatement` SELECT | same | Parameterized SELECT |
| 57 | 12 | JDBC INSERT/UPDATE/DELETE | same | No string-concat SQL |
| 58 | 12 | JDBC Library CRUD mini + injection note | same | Push JDBC CRUD |

## Phase D — Frontend (Sessions F1–F9, 63–65)

**HTML & CSS (F1–F9):** nine sessions from the HTML & CSS learning path, one AcmeHR page per session. They are numbered F1–F9 so that every other session keeps its number; week **12A** is the extra bridge week.

| S# | Wk | Focus | Module | Homework |
|---|---|---|---|---|
| F1 | 12 | How the web works & first page | Lesson_01 · lab | Welcome page served locally |
| F2 | 12 | Content & first CSS | Lesson_02 · lab | Leave policy page |
| F3 | 12A | Page structure & box model | Lesson_03 · lab | Employee profile |
| F4 | 12A | Images, navigation & Flexbox | Lesson_04 · lab | Employee directory |
| F5 | 12A | Grid & positioning | Lesson_05 · lab | HR dashboard |
| F6 | 12A | Forms (GET/POST, what the server receives) | Lesson_06 · lab | Apply-for-leave form |
| F7 | 12A | Tables | Lesson_07 · lab | Attendance report |
| F8 | 13 | Responsive design | Lesson_08 · lab | Dashboard on a phone |
| F9 | 13 | Polish, accessibility & final project | Lesson_09 · lab | Final project: AcmeHR portal |
| 63 | 13 | JS variables, DOM, events | ../FRONTEND/03_JavaScript_DOM_Forms.md | Click counter |
| 64 | 13 | Form validation | same | Validate contact |
| 65 | 13 | `fetch` JSON overview | same | Portfolio page push |

## Phase E — Spring Boot (Sessions 66–80)

| S# | Wk | Focus | Module | Homework |
|---|---|---|---|---|
| 66 | 14 | Maven + Spring Boot init | 09_Spring_Boot/01_Maven_and_Spring_Boot_Setup.md | `mvn spring-boot:run` proof |
| 67 | 14 | `application.yml` + first `@RestController` | same | Hello JSON |
| 68 | 14 | Mappings, JSON, status codes | 09_Spring_Boot/02_Spring_Web_REST.md | GET list (in-memory) |
| 69 | 14 | `@PathVariable` / `@RequestBody` | same | POST + GET by id |
| 70 | 14 | Thymeleaf **overview only** (one page) | same | Optional HTML view note |
| 71 | 15 | JPA entity + repository | 09_Spring_Boot/03_Spring_Data_JPA_Validation.md | Product entity |
| 72 | 15 | JPA CRUD + MySQL | same | Persist catalog |
| 73 | 15 | Validation + DTO mapping | 09_Spring_Boot/04_REST_DTOs_Postman.md | `@Valid` on POST |
| 74 | 15 | Postman collection + error bodies | same | Collection v1 |
| 75 | 15 | Security lite: register/login (JWT) | 09_Spring_Boot/05_Spring_Security_Lite.md | Auth works |
| 76 | 16 | **Lab 06** CRUD/auth consolidation | 09_Spring_Boot/06_Lab_CRUD_Auth_Consolidation.md | Feature freeze list |
| 77 | 16 | **Lab 06** continue + peer break-auth | same | Must-pass items |
| 78 | 16 | **Lab 06** demos | same | Deliverables bundle |
| 79 | 16 | **Lab 07** deploy debug | 09_Spring_Boot/07_Lab_Deploy_Capstone_Debug.md | Actuator / health + env |
| 80 | 16 | **Lab 07** PaaS attempt + capstone M0 | Capstone_CampusCart_One_Pager.md | Auth shell push |

## Phase F — Interview map, React, Quality, Capstone (Sessions 81–100)

| S# | Wk | Focus | Module | Homework |
|---|---|---|---|---|
| 81 | 17 | Servlet vs Spring (interview map only) | 09_Spring_Boot/08_Servlet_vs_Spring_Interview_Map.md | 5 comparison bullets |
| 82 | 17 | CORS + API recap for React | 09_Spring_Boot/04_REST_DTOs_Postman.md | CORS config note |

| S# | Wk | Focus | Module | Homework |
|---|---|---|---|---|
| 83 | 17 | Capstone M1 auth shell | Capstone one-pager | Login/register API |
| 84 | 17 | Capstone M2 catalog JPA | same | Product CRUD |
| 85 | 17 | Capstone M3 cart/orders start | same | Order flow stub |
| 86 | 18 | React Vite + first component | ../FRONTEND/REACT/01_React_Fundamentals.md | Hello card |
| 87 | 18 | Props + state | same | Counter + list |
| 88 | 18 | Forms + lists/keys | same | Controlled form |
| 89 | 18 | `useEffect` + Router | ../FRONTEND/REACT/02_React_Hooks_Routing_API_Deploy.md | Two routes |
| 90 | 18 | Fetch Spring API + CORS | same | Product list UI |
| 91 | 19 | Debugging + logging habits | 12_Testing_Clean_Code/01_Testing_Debugging_Clean_Code.md | Debugger lab |
| 92 | 19 | JUnit 5 + Mockito lite | same | 3 tests |
| 93 | 19 | Capstone M2–M3 features | Capstone one-pager | Orders flow |
| 94 | 19 | Capstone polish + a11y pass | Deliverables + one-pager | Screenshots |
| 95 | 19 | Deploy capstone / Lab 07 catch-up | Capstone one-pager | Live or staging URL |
| 96 | 20 | Resume + LinkedIn + GitHub pins | 13_Placement/01_Career_Resume_Interview.md | Resume v1 |
| 97 | 20 | Mock interview (Java + SQL) | same | Feedback notes |
| 98 | 20 | Mock interview (Spring + projects) | same | STAR stories |
| 99 | 20 | Capstone defense rehearsals | Capstone | 10-min script |
| 100 | 20 | Demo day + placement wrap | Placement | Applications started |

**AI Integration (12_AI_Integration):** optional after S95 — never displaces S93–S100.

## If the batch falls behind

**Slip React / JWT depth / Thymeleaf**, not Spring Lab 06/07, JDBC `PreparedStatement`, or SQL joins. Those are placement-critical.